

RC Primer: Incredibly Easy Parallelization with High Throughput Computing

Research Computing
UNIVERSITY OF COLORADO BOULDER

Incredibly Easy Parallelization with High Throughput Computing

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- Website: www.rc.colorado.edu/rc
- Documentation: https://curc.readthedocs.io
- Helpdesk: rc-help@colorado.edu
- Survey: http://tinyurl.com/curc-survey18



Slides
https://github.com/ResearchComputing/easy_parallelization_htc_primer

Meet the User Support Team



Layla
Freeborn



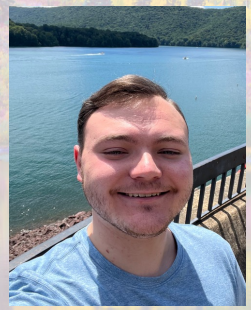
Brandon
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Andy
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Michael
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John
Reiland



Dylan
Gottlieb



Mohal
Khandelwal



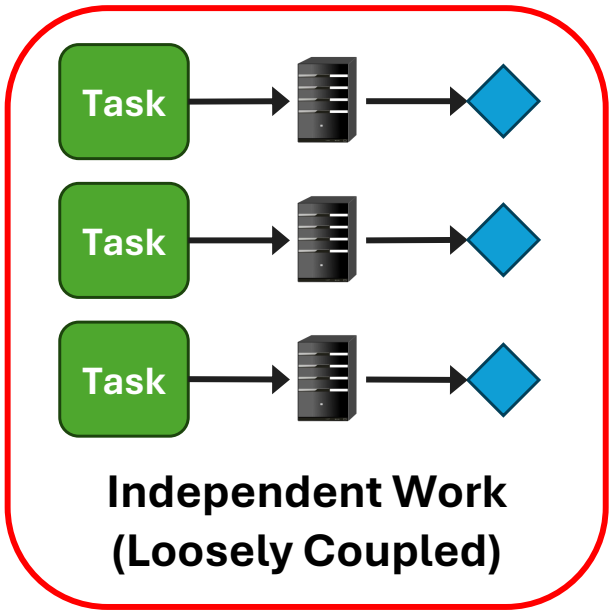
Ragan
Lee

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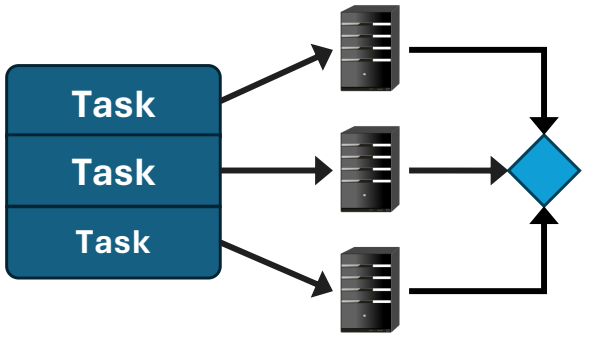
Learning Objectives and Outline

- Introduction to high throughput computing (HTC)
- HTC Tools
- Structuring Data for HTC
- Open Science Grid

What is High Throughput Computing (HTC)?



Independent Work
(Loosely Coupled)



Dependent Work
(Highly Coupled)

This is HTC!

Concept Review – Is It High Throughput?

Workflow A

Analyze a collection of medical records to identify patients most at risk for heart disease.



Workflow B

Analyze atmospheric data to simulate and forecast weather conditions over a large geographic area.



Starter Script – Low Throughput

CODE

```
import sys
car=sys.argv[1]
mpg=sys.argv[2]
print("The " + car + " gets " + mpg + " mpg.")
```

DATA

```
mustang 25
pinto 30
chevette 33
nova 21
cutlass 23
```

```
The mustang gets 25 mpg.
The pinto gets 30 mpg.
[…]
```

Does it Scale?

If 1 task takes 30 minutes to finish, how well does the current script scale?

5 tasks?

5,000 tasks?

5 Million tasks?

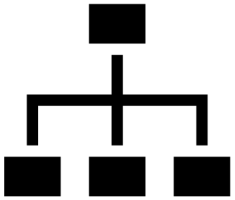
Three HTC Methods

Slurm Job Arrays



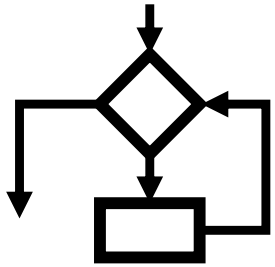
Scaling Up
Big Jobs

Load Balancer



Built-In
Resource Manager

GNU Parallel



Replaces
For-Loops

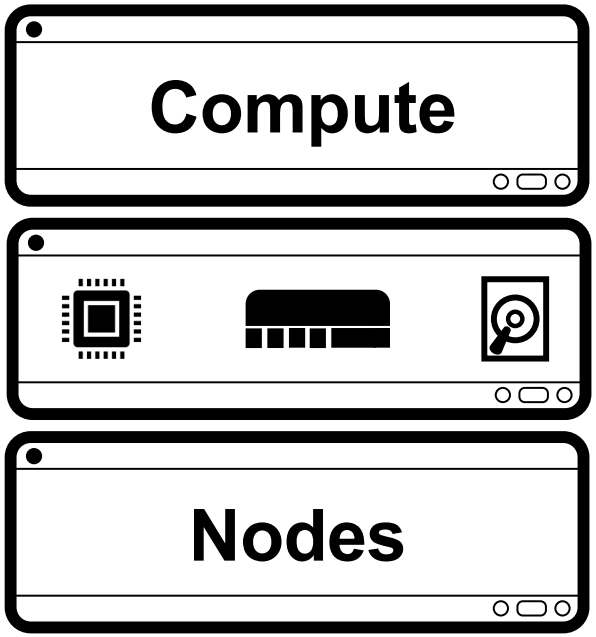
What is a Batch Job?

```
#!/bin/bash

#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --time=00:10:00
#SBATCH --partition=atesting
#SBATCH --qos=testing
#SBATCH --output=sample-%j.out
…
```

Batch Job

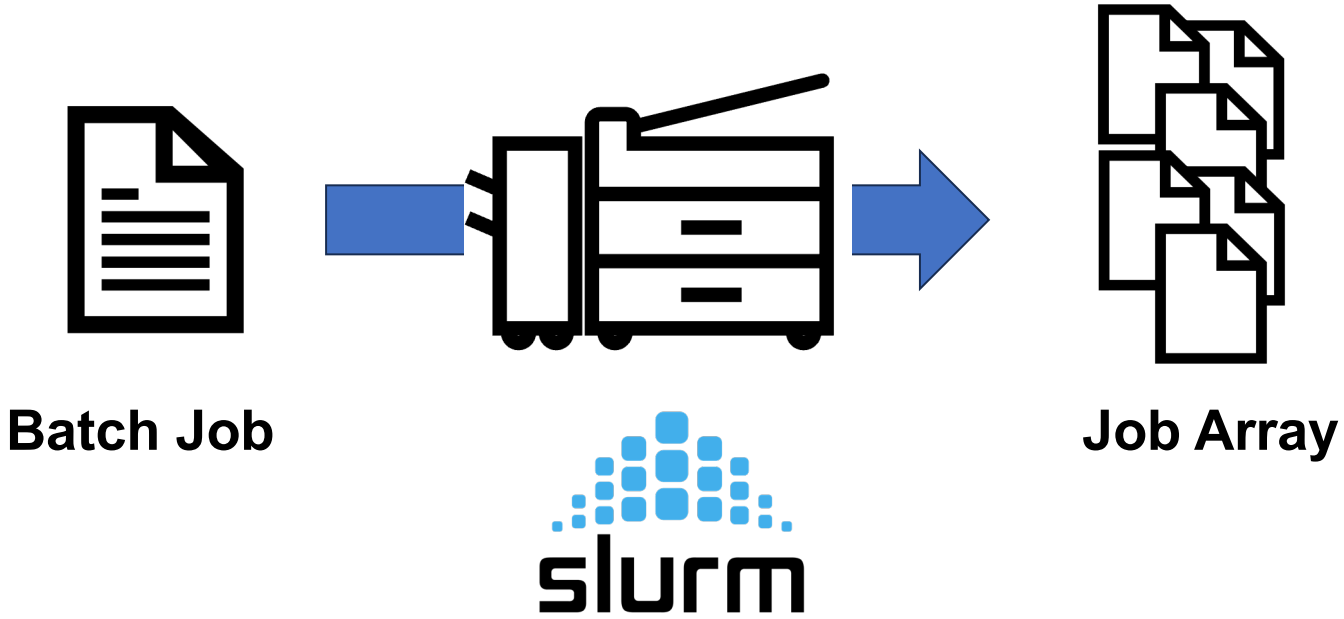




SLURM Job Array



Slurm Job Array - Overview



SLURM Job Arrays - Implementation

cars_mpg_array.sh (job script)

```
#!/bin/bash
#SBATCH --time=00:00:10
#SBATCH --partition=amilan
#SBATCH --qos=normal
#SBATCH --nodes=1
#SBATCH --ntasks=1
#SBATCH --job-name=cars
#SBATCH --output=cars.%A_%a.out
#SBATCH --array=1-5

# load anaconda python
source /curc/sw/anaconda/default

# run workflow
python cars_mpg.py $(sed -n "${SLURM_ARRAY_TASK_ID}p" cars_mpg_input_args_array.txt)
```

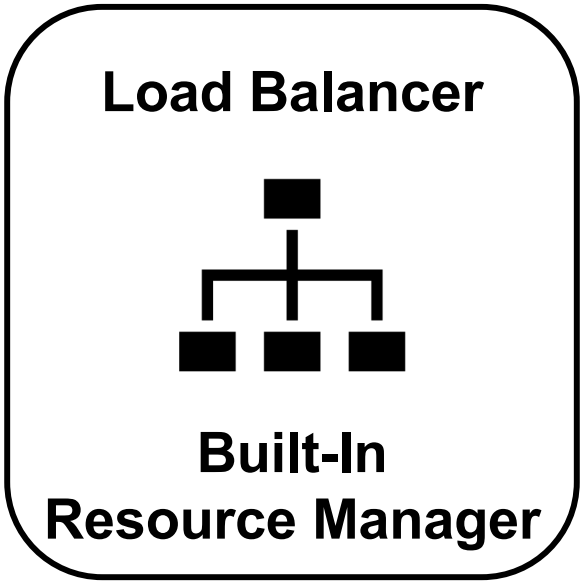
```
#SBATCH --array=START-STOP
#SBATCH --ntasks=Cores per
#SBATCH --nodes=Nodes per
```

cars_mpg_args_array.txt (arguments)

```
mustang 25
pinto 30
chevette 33
nova 21
cutlass 23
```

$SLURM_ARRAY_TASK_ID → 1 - 5

Load Balancer



Method 2 Example: Load Balancer

cars_mpg_lb.sh (job script)

```
#!/bin/bash
#SBATCH --time=00:00:10
#SBATCH --partition=amilan
#SBATCH --qos=normal
#SBATCH --nodes=1   # optional
#SBATCH --ntasks=6
#SBATCH --job-name=cars
#SBATCH --output=cars.%j.out


# load loadbalancer and anaconda python
module purge
module load loadbalance
source /curc/sw/anaconda/default

# run workflow
mpirun lb cars_mpg_input_commands_lb.txt
```

Request resources for all members

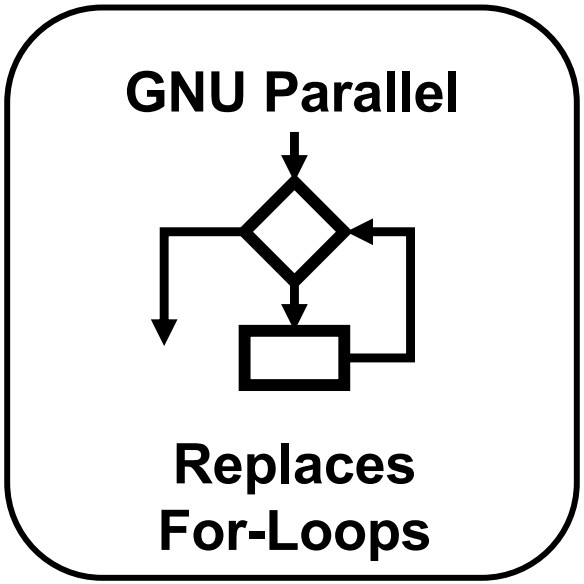
Load the 'loadbalance' module

Workers = ntasks - 1

cars_mpg_commands_lb.txt (arguments)

```
python cars_mpg.py mustang 25
python cars_mpg.py pinto 30
python cars_mpg.py chevette 33
python cars_mpg.py nova 21
python cars_mpg.py cutlass 23
```

GNU Parallel



Method 3 Example: GNU Parallel

cars_mpg_gnuparallel.sh (job script)

```
#!/bin/bash
#SBATCH --time=00:00:10
#SBATCH --partition=amilan
#SBATCH --qos=normal
#SBATCH --nodes=1   # optional
#SBATCH --ntasks=5
#SBATCH --job-name=cars
#SBATCH --output=cars.%j.out


# load gnu parallel and anaconda python
module purge
module load gnu_parallel
source /curc/sw/anaconda/default

# run workflow
parallel -j $SLURM_NTASKS < cars_mpg_input_commands_gnuparallel.txt
```

Request resources for all members

Load the 'gnu_parallel' module

Workers = ntasks

Tasks Split Evenly

cars_mpg_commands_gnuparallel.txt (arguments)

```
python cars_mpg.py mustang 25
python cars_mpg.py pinto 30
python cars_mpg.py chevette 33
python cars_mpg.py nova 21
python cars_mpg.py cutlass 23
```

Three HTC methods for Alpine

	Job Arrays	Load Balancer	GNU Parallel
Job length?	> 10 min	< 30 min	< 30 min

Three HTC methods for Alpine

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Where can I use?	Alpine, SLURM HPC	Alpine	Alpine, Local Computer

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Max jobs in queue?	1000*	-	-

*Depends on QOS

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Max jobs in queue?	1000*	-	-
Supports Multi-Node?	Yes	Yes	Yes – but not well

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Does it reserve a controller core?	No	Yes	No

*Depends on QOS

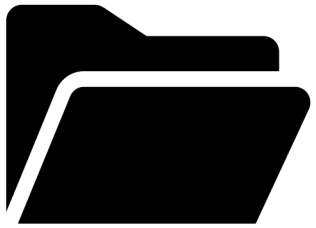
Three HTC methods for Alpine

	Job Arrays	Load Balancer	GNU Parallel
Job length?	> 10 min	< 30 min	< 30 min
Where can I use?	Alpine, SLURM HPC	Alpine	Alpine, Local Computer
Max jobs in queue?	1000*	-	-
Supports Multi-Node?	Yes	Yes	Yes – but not well
Max cores per task?	None	None – Usually 1	None – Usually 1
Does it reserve a controller core?	No	Yes	No
Primary Focus?	Managing jobs	Manages Resources Across Nodes (MPI-Lite)	Replaces Loops

*Depends on QOS

Structuring Data

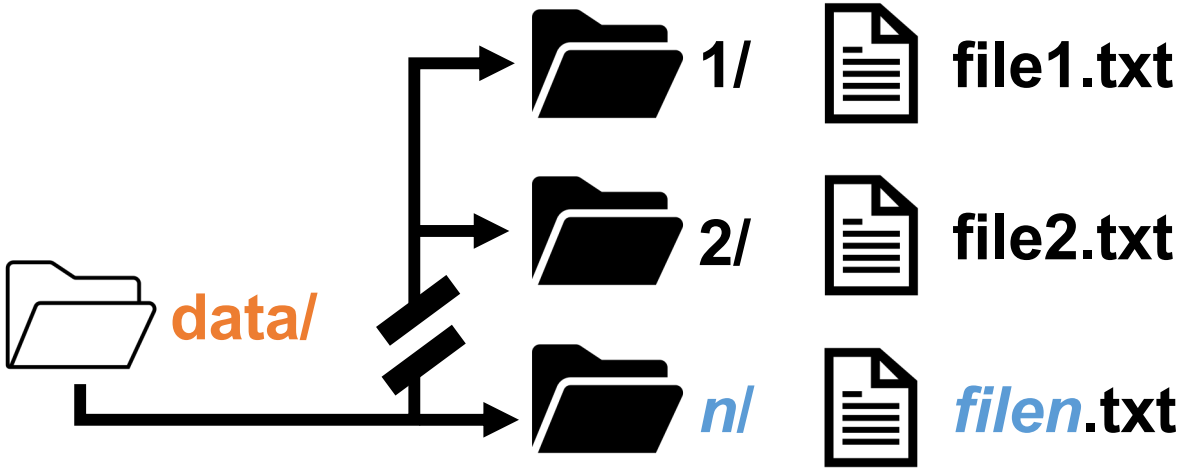
Files & Directories



Inputs & Parameters



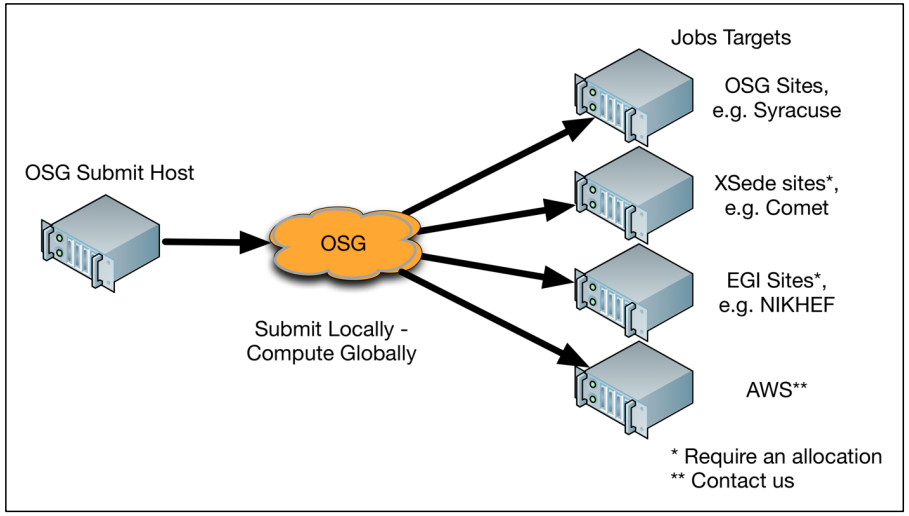
Structuring Files & Directories



/scratch/alpine/$USER/data/*n*/file*n*.txt

A Community HTC Resource beyond Alpine: Open Science Grid (OSG)

- NSF/DOE-funded service (free!)
- Open to any U.S.-based researcher
- 125 institutions sharing spare computing cycles
- ~1 Billion core hours per year used
- Can get started quickly. Options for dedicated allocation if needed.
- https://osgconnect.net



Source: https://swc-osg-workshop.github.io/OSG-UserTraining-JLab-2019/materials/AHM/01-IntroGrid.html

Thank you!

Survey and feedback

http://tinyurl.com/curc-survey18

